

Lesson 2.2 — Deploy to Vercel

Trainer's Speech & Lecture Script

Duration: ~18 minutes | Day 2 of 5 | Build Your Own AI Web Application

HOW TO USE THIS SCRIPT

Text in **[INSTRUCTOR ACTION]** = what you DO on screen, not what you say.

Text without brackets = what you SAY out loud to the class.

Pause after each section and check that students are following before moving on.

If a student is stuck on deploy: stop the class. Do not leave anyone behind — they will be blocked for the rest of the course.

OPENING HOOK

(~1 minute)

"Before we do anything else, I want you to think about this.

Right now, your app lives on your laptop. That means only YOU can see it. If I ask you to show me your app, you have to be sitting next to your machine. If your laptop dies, the app dies.

An app on your laptop is a science project. An app with a public URL is something a stranger can actually use — and a judge can actually open.

Today we fix that.

By the end of this lesson, every one of you will have a live URL on the internet that anyone in this room can open on their phone right now. That is today's goal. One live URL. Let's go."

SECTION 1 — WHY WE DEPLOY TODAY, NOT DAY 5

(~3 minutes)

"You might be wondering — why are we deploying NOW? We haven't added AI yet. Why not wait until we're done?

Here is the answer: **the first deploy is always the most painful one.**

There are three specific things that break on first deploy almost every single time. I call them the three deploy walls.

We are going to hit all three today, on Day 2, while it's low-stakes. Nothing critical is on the line yet.

If we wait until Day 5 and hit these walls at the deadline, you're in trouble. You cannot submit a broken URL. So we do the painful part now, while it's cheap.

Also — once your GitHub repo is connected to Vercel, every time you push code, Vercel automatically rebuilds and redeploys your site.

Push code. Site updates. That simple."

SECTION 2 — WHAT IS VERCEL

(~2 minutes)

"What is Vercel?

Vercel is a hosting platform. You give it your GitHub repo, it reads your code, builds it, and gives you a public URL that anyone can visit.

It knows how to run Next.js apps perfectly — it was made by the same people who built Next.js. It's free for personal projects. No credit card needed.

Think of it this way:

- **GitHub** is where your CODE lives — the filing cabinet
- **Vercel** is where your APP runs — the shop front

GitHub stores it. Vercel runs it. They are connected — Vercel watches your GitHub repo, and any time you push a change, it automatically redeploys.

Clear? Good. Let's build."

SECTION 3 — STEP BY STEP: DEPLOYING TO VERCEL

(~5 minutes)

STEP 1 — Push your latest code to GitHub

[INSTRUCTOR ACTION: Open terminal on screen]

"First, make sure your latest code is on GitHub. Open your terminal and run these three commands:"

```
git add .
git commit -m "ready for deploy"
git push
```

"If you see any errors — stop. Raise your hand. Do not move forward until your code is on GitHub.

Now go to github.com, find your repo, and refresh the page. You should see your latest files. That is your confirmation."

Pause — wait for students to confirm their code is on GitHub.

STEP 2 — Create a Vercel account

[INSTRUCTOR ACTION: Open vercel.com in browser]

"Open vercel.com. Click Sign Up with your GitHub account — this is important. It's free. No credit card. Once logged in, you will see the Vercel dashboard."

Pause — wait for students to log in to Vercel.

STEP 3 — Import your repository

[INSTRUCTOR ACTION: Click 'Add New Project' in Vercel dashboard]

"Click 'Add New Project'. Vercel will show your GitHub repos. Find your project and click Import.

On the next screen: **do NOT change the framework preset** — Vercel detects Next.js automatically.

BUT — before you click Deploy — **STOP.**

This is where beginners get burned. We need to do two things first. If you click Deploy right now your app will break and you will spend an hour confused.

Everyone: hands off the Deploy button. Follow me."

SECTION 4 — THE THREE DEPLOY WALLS

(~8 minutes — the most important part of today)

"Almost every beginner hits the same three walls on first deploy. I am going to walk you through all three right now. Pay close attention."

WALL 1 — ENVIRONMENT VARIABLES

(~3 minutes)

"Remember the file on your machine called **.env.local**?

That file holds your MONGODB_URI — the secret connection string for your database.

Here is the problem: .env.local is in your .gitignore. It was NEVER pushed to GitHub. Vercel has

never seen it. When Vercel builds your app and tries to connect to MongoDB, it finds nothing. Your app crashes immediately.

The fix: you must manually tell Vercel your environment variables.”

[INSTRUCTOR ACTION: Show Environment Variables section in Vercel project settings]

Find the 'Environment Variables' section and add:

```
Name: MONGODB_URI
Value: mongodb+srv://yourname:yourpassword@cluster0.abc123.mongodb.net/yourappname
```

“Paste the full string. No spaces before or after. No quote marks. Just the raw string exactly as it appears in your .env.local file.

Rule for the rest of your career:

NEVER put your connection string in your code.

NEVER put it in your README.

NEVER commit it to GitHub.

Secrets live in .env.local on your machine and in Vercel’s environment variable panel in the cloud. Nowhere else.”

Pause — check every student has entered MONGODB_URI in Vercel.

WALL 2 — MONGODB ATLAS NETWORK ACCESS

(~3 minutes)

“Here is the second wall. This one is sneaky — your app will BUILD successfully, Vercel will show a green checkmark — but when you actually try to USE your app, it times out.

The reason: MongoDB Atlas only allows connections from your home IP address by default.

Vercel’s servers are in a data center with completely different IP addresses. Atlas sees an unknown IP and blocks it.

The fix: tell Atlas to allow connections from anywhere.”

[INSTRUCTOR ACTION: Open MongoDB Atlas → Network Access in left sidebar]

“Go to mongodb.com, log in, open your project, click **Network Access** in the left sidebar.

Click **Add IP Address**.

Click **Allow Access from Anywhere**. It fills in:

0.0.0.0/0

That means: all IP addresses, everywhere.

Click Confirm.

For a real production app you would restrict this later. But for this course, if you do NOT do this, your deployed app will never connect to the database. Open it up.”

Pause — check every student has added 0.0.0.0/0 in Atlas Network Access.

WALL 3 — EXACT CONNECTION STRING

(~2 minutes)

“The third wall is simple but it trips people up every time. Your connection string must be exactly right.

Common mistakes:

- A typo anywhere in the string
- Extra space at the beginning or end when pasting
- Wrong database name at the end
- Using an old password you changed later

Right now: go to MongoDB Atlas → click your cluster → Connect → Connect your application → copy the string fresh from there.

Replace USERNAME and PASSWORD with your real credentials. Paste it fresh into Vercel.

Do not use a string you typed by hand. Always copy it directly from Atlas.”

Pause — check students have verified and re-entered their connection string.

SECTION 5 — CLICK DEPLOY AND WATCH THE BUILD

(~2 minutes)

“Now. We have:

- Code pushed to GitHub
- MONGODB_URI entered in Vercel environment variables
- MongoDB Atlas Network Access open to 0.0.0.0/0
- Connection string verified and exact

NOW you can click Deploy.”

[INSTRUCTOR ACTION: Click the Deploy button in Vercel]

“Vercel will build your project. You will see a build log scrolling down the screen. This takes about 30 to 60 seconds.

Watch for errors. If it fails — look at the last few lines of the build log. The error will be there. Read it

out loud or raise your hand.

If it succeeds — you will see a green checkmark and a live URL.

That URL is your app. On the internet. Right now.

But — the build being green does NOT mean the app works. We have one more step.”

Pause — wait for all builds to complete.

SECTION 6 — TEST LIKE A STRANGER

(~2 minutes)

“This is the most important test you will do today.

Open your live URL in a **fresh incognito window**. Not a regular tab. Incognito. Pretend to be a complete stranger who has never seen this app.

Now do the core loop:

1. Create an item — add a piece of data using your form
2. Reload the page
3. Confirm the item is still there

If the item survived a page reload, your data is being saved to MongoDB Atlas. That is real persistence. That is your app working for real.

If your app loads but adding data fails: almost certainly Wall 2. Go back to Atlas Network Access and confirm 0.0.0.0/0 is saved.

If your app does not load at all: almost certainly Wall 1. Go back to Vercel environment variables and check MONGODB_URI.”

SECTION 7 — DISCUSSION PROMPT

(~2 minutes)

“Now we are going to do something fun.

Everyone who has a working live URL — turn to the person next to you and swap URLs.

Open their URL in your browser. Add an item to their app. They add an item to yours. Then reload. Is their data still there? Is yours?

This is what ‘deployed’ means. A stranger just used your app. That is a real thing that exists on the

internet right now.”

Give students 2-3 minutes to swap URLs and test each other's apps.

SECTION 8 — KNOWLEDGE CHECK (Quiz the Room)

(~2 minutes) Ask these questions out loud. Students call out the answers.

Q1: Why do we deploy on Day 2 instead of waiting until Day 5?

Answer: Deploy config is required today, and the first deploy is always fiddly — fight it while it's low-stakes, not at the deadline.

Q2: Once your GitHub repo is connected to Vercel, what happens every time you push?

Answer: Vercel automatically rebuilds and redeploys the live site. Push code, site updates.

Q3: Your deployed app times out connecting to MongoDB but works locally. Most likely cause?

Answer: Atlas Network Access doesn't allow Vercel's servers. Add 0.0.0.0/0 in Atlas Network Access and confirm MONGODB_URI env var is set in Vercel.

Q4: How do you properly test that your deploy actually works?

Answer: Open the public URL in a fresh incognito browser and run the core loop: add an item, reload, confirm it survived. A green build is not enough.

SECTION 9 — RECAP AND CLOSE

(~1 minute)

“Let me summarise what you did today.

You took an app that only existed on your laptop and put it on the internet.

You learned the three deploy walls every beginner hits:

Wall 1 — Env vars must be re-entered in Vercel manually

Wall 2 — Atlas Network Access must be opened to 0.0.0.0/0

Wall 3 — Connection string must be exactly right — copy fresh from Atlas

You know that a green Vercel build does not mean your app works. You must test as a stranger in incognito.

And from now on, every git push you make automatically redeploys your live site.

Watch the recap video at the end of this lesson. You must watch it through to complete the lesson and unlock tomorrow.

Tomorrow — Day 3 — we add the AI. We make your app talk to a language model. See you then.”

QUICK REFERENCE CARD — STICK ON THE WALL

THE THREE DEPLOY WALLS	
Wall 1: Env Vars	Vercel → Settings → Environment Variables Add: MONGODB_URI = [full Atlas connection string]
Wall 2: Atlas Network Access	Atlas → Network Access → Add IP Address Add: 0.0.0.0/0 (Allow Access from Anywhere)
Wall 3: Exact String	Atlas → Connect → Connect your application → copy fresh Replace USERNAME and PASSWORD with real credentials
How to Test	Open live URL in incognito → add item → reload → confirm it's there
Auto Deploy	Every git push to main → Vercel rebuilds → live site updates
.env.local Rule	NEVER commit to GitHub. NEVER write in code. Secrets: .env.local (local) + Vercel env vars (cloud). Nowhere else.